VIRTUAL REALITY FOR EMOTIONAL REGULATION IN ACADEMIC SITUATIONS: USABILITY AND USER EXPERIENCE EVALUATION

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Abstract:

Currently, the use of information technologies, especially virtual reality, to improve learning experiences in education has become a trend in Higher Education. This study focuses on the implementation for the analysis of usability and user experience of the 'EmotionLab' prototype, using Oculus Quest 2 glasses, designed for emotional regulation during academic situations. In methodological terms, a quasi-experimental cross-sectional study was conducted, involving 12 Higher Education students, who were considered as case studies and were administered the short version of the UEQ questionnaire, as well as key indicators such as the Net Promoter Score, error rate and dropout rate. Preliminary findings indicate that students found the experience innovative, entertaining and helpful in improving their emotion management during oral presentations. No signs of 'cybersickness' were reported, and no dropouts were reported during the experience. This study suggests that virtual reality as a pedagogical tool has great potential in Higher Education to promote active and emotionally balanced learning.

Keywords: *Virtual Reality; User Experience; Educational Technology; Emotional Regulation; Higher Education.*

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Introduction and conceptual framework

After experiencing disruptive events such as the global pandemic and other factors of modern life, students find themselves in a context in which they need greater emotional support to enhance their academic and personal development. In this sense, there is a tendency towards a propensity to frustration and a diminished confidence in their own abilities. (Martínez et al., 2021). These emotional challenges have a direct impact on their ability to engage and progress in the educational process. It is for this reason that the student is recognized not only as a thinking being, but also as the protagonist of his own learning and that, through his thoughts, actions and feelings, he promotes and achieves lasting learning. It is understood then, that the emotional dimension significantly affects the acquisition and permanence of learning, either positively or negatively.

Based on the above, the understanding of emotions, both by educators and students themselves, has become an essential component to promote valuable, authentic and enriching learning. In the educational field, there has been a transition towards innovations in pedagogical strategies for the achievement of meaningful learning, which translates into the development of skills that allow solving complex tasks related to the world of work, so that it is no longer enough to transmit knowledge in a unidirectional manner. (Matienzo, 2020). It is now understood that learning goes beyond the accumulation of information; it is a holistic process that involves both the mind and the heart (Bueno, 2021; Davalillo, 2014). In this context, the socioemotional connection of the student emerges as a fundamental element in the promotion of a healthy educational environment conducive to authentic learning. Consequently, it is necessary to take care of the socioemotional connections in the educational environment, which implies recognizing and attending to the emotional needs of the students. In this sense, the specialized literature proposes that emotion is a central element in the learning process; when students feel motivated and emotionally connected to what they are learning, the process becomes more meaningful and lasting (Bisquerra, 2009; Bueno, 2021), which supports that the incorporation of strategies that consider emotional management can transform the pedagogical practice and improve the motivation and academic success of students (Salcedo de la Fuente et al., 2024).

Currently, digital transformation, the irruption of the metaverse and digital devices are increasingly integrated into the teaching-learning process in Higher Education (Molina, 2024), modifying the way in which teaching and learning take place, providing new opportunities for the acquisition of knowledge, skills and competencias (Ramos & Peredo, 2023). Within the emerging technologies that are being integrated in a disruptive way in Higher Education, which are transforming and improving the teaching - learning process, are the extended technologies, a term that refers to augmented reality, mixed reality and virtual reality, which are being incorporated within the educational, research and management process of tertiary education institutions, generating a space for innovation and experimentation (Cabero & Fernández, 2018).

In this context and for the purposes of this study, virtual reality is a technological tool that is being used for different solutions such as health, sports and education (Agbo et al., 2023; Ammann-Reiffer et al., 2022; Neher et al., 2024; Zhang et al., 2020). As far as education is concerned, the use of this technology is transforming and improving the teaching - learning process, allowing to provide

students with an immersive and enriching learning experience, which allows them to experience and practice academic situations in a functional way and with more meaningful learning (Cabero & Fernández, 2018). In turn, the integration of virtual reality based on serious games in Higher Education is reshaping the way students learn, so, it invites to reflect on the learning methodologies with which professors teach, which presents a very promising future, further enhanced, by the wide availability in the market of software and hardware tools, which leads to a significant improvement in graphics, with increasingly realistic experiences and which in turn are becoming more affordable for the development of virtual environments, research and experimentation (Checa & Bustillo, 2020).

With reference to learning experiences in Higher Education, these have long followed a traditional approach, where predefined formats are established and applied in physical teaching environments (Gallardo Fernández et al., 2020). However, currently technology has significantly burst into the educational field, transforming the way in which knowledge is acquired. This transformation has led to a paradigm shift in the learning design and planning process, which is now strongly influenced by technology, especially with regard to the design of educational experiences, which establishes that the teaching-learning process must adapt to the use of technology (Checa & Bustillo, 2020).

However, immersive learning experiences, while offering great potential for experiential learning, present unique challenges in relation to cognitive load. Cognitive load is defined, according to, Paas et al., (2003) as the amount of mental resources a person brings to bear when performing a task in a specific environment, which can be affected by the complexity of the virtual environment, the user interface, and the demands of the task. Therefore, an overly complex immersive learning experience, with excessive visual and auditory information, can overload the learner's working memory, leading to decreased comprehension, disorientation and retention of information, making it a critical aspect that requires attention in its implementation (Cabero-Almenara et al., 2023).

In order to attenuate the cognitive load in immersive learning experiences, designers should consider instructional design principles based on scientific research, such as spatial coherence, which suggests that visual and auditory information should be presented close to its spatial referent in the virtual environment, reducing the need for visual search and decreasing the load on working memory. Similarly, verbal information is best presented auditorily, while visual information is presented with images, this in order to take advantage of the different information processing channels available in the brain (Sweller et al., 2019).

The design of immersive learning experiences has a profound impact on cognitive processes. In this line of argument, Sjölie, (2012) argues that the human brain processes experiences in virtual environments similarly to those in the real world, creating mental simulations regardless of the nature of the environment. This means that the design of an immersive learning experience can influence the user's attention, memory, spatial perception, and even emotional responses (Neroni et al., 2021)

In this sense, designers of this type of learning experiences should consider some relevant aspects such as the user's previous experience to determine the complexity of the environment and

the pace of information presentation. Inexperienced users or users with no previous experience in immersive environments can benefit from simpler environments and more explicit guidance, while users with previous experience can handle more complex environments and a higher degree of autonomy. Therefore, carefully addressing cognitive load in the design of virtual environments will create more effective and engaging learning experiences for learners (Palacios-Rodríguez et al., 2024).

Therefore, the teaching-learning process is moving towards immersive learning experiences that comprehensively involve intellectual, affective, social and physical aspects (Sánchez Castillo, 2023). This type of modality allows generating experiences that use extended technologies such as virtual, augmented or mixed reality considered complex, but using a didactic plan integrating new and innovative pedagogical and didactic strategies such as role-playing, situated learning, problem based learning, among others (Torres & Rodríguez, 2019).

The aforementioned facilitates experiencing scenarios that are difficult to access or complicated contexts, expanding aspects of reality that are difficult to experience, enabling a context and a sense of presence. In other words, immersive learning experiences support active learning, enhance the depth of learning and stimulate interactions between students and the virtual environment (Agbo et al., 2023). In addition, they offer an educational experience that considers the characteristics of the current entertainment industry, such as interactivity, animation, color, sound, but with a didactic plan and a specific pedagogical strategy. However, immersive learning experiences with virtual reality have not adequately integrated learning theories in their design and development, which represents an opportunity to incorporate these theories in the creation of virtual reality applications that promote more effective learning.

In the same way, people interact more frequently with different technological devices such as computers, cell phones, sensors, HMD glasses, among others, so it is important to take into account both the user experience and usability in the design of these immersive learning experiences with virtual reality. These concepts are fundamental for the acceptance of these experiences, as they focus on the effective and pleasurable interaction of users with virtual environments and which are also related to human factors, where the International Organization for Standardization (ISO 9241 11:2018), defines them as follows: a) on the one hand, user experience refers to the perceptions and responses of a person from the use of the product and b) usability is the extent to which a virtual reality experience can be used to achieve the objectives effectively, efficiently and with satisfaction in a specific context of use. However, as virtual reality evolves, user experience design adapts to improve usability, ensuring that users can navigate, interact with and enjoy virtual experiences intuitively and realistically.

The research that supports this article is based on an immersive learning experience called "EmotionLab" that is currently being developed at the Technology Center (CeT) of the School of Computer Science and Telecommunications of DuocUC, Viña del Mar Campus, in which technology such as Immersive Virtual Reality (HMD) is used for learning, based on active pedagogical strategies such as Problem-Based Learning (PBL) and Game-Based Learning (GBL). In addition, the experience explores the crucial role of immersive technology in Higher Education, especially on the role of

emotion management and its impact on the teaching-learning process. In this sense, (Lerner et al., 2010) indicates that the behaviors manifested arise from regulatory processes and can serve as predictors of future functioning and performance, thus substantiating that emotional regulation is a relevant issue in any area of life. For the regulation of emotions, there are a variety of models to optimize learning. For this research it is proposed to use Gross and Thompson's model of emotional regulation for learning in (Saavedra et al., 2015). This model considers four components in the emotion generation process: a) a relevant situation b) attentional processes directed to the situation c) thoughts about it and d) finally the emotional response. This model will be the one proposed for the design of learning experiences with immersive virtual reality, since it will allow us to understand the emotions and be able to organize how to regulate them in each of the components being able to change the situation, try to change the attentional focus, modify the thought by reevaluating the situation and change the expressive response this will allow to improve the expressive response, decrease the physiological activity, such as heart rate, which will improve the memory and greater contact with other people. Therefore, the main objective of this study is to investigate the usability, user experience and acceptability of an immersive learning experience called "EmotionLab", through Oculus Quest 2 (HMD) virtual reality glasses designed to promote emotional regulation for cognitive learning in higher education students.

Methodology

Design

The present study is supported by a post-positivist research paradigm, which assumes that the observed reality cannot be captured in its entirety, as assumed by classical positivism, as well as recognizing the subjectivity of researchers and study subjects in the process(Creswell et al., 2018; Yin, 2018). In turn, the design of this research is quasi-experimental, single test in each of the cases, considering these subjects as cases of multiple studies (Yin, 2018), in which data of a quantitative nature were obtained which were analyzed using tools of descriptive and inferential statistics.

Participants

For the purposes of this study, a non-probabilistic purposive sampling was developed and according to the availability of the students (Creswell et al., 2018; Flick, 2023) which was executed in a controlled context in the facilities of the Higher Education institution to which the subjects belong. The participants corresponded to students, both female and male, between 18 and 25 years old, belonging to the careers of Physical Trainer, Digital Animation, Graphic Design and Computer Science of the institution DuocUC, Sede Viña del Mar.

The objective of the sampling process was to have a group of students who were required to give an individual oral presentation as part of the testing of the virtual reality prototype designed for HMD Oculus Quest 2 goggles.

Before proceeding to the selection of the participants, the inclusion and exclusion criteria for this usability and user experience test were defined, which in this case considered higher education students specifically enrolled at DUOC UC, Viña del Mar campus. Both women and men were included, between the ages of 18 and 25 years old, belonging to the daytime and who were in the process of taking their final degree exam. No distinction was made between those with or without previous experience in the use of virtual reality. On the other hand, the exclusion criteria considered those individuals outside the context of higher education, students who do not belong to the DUOC UC institution, those under 18 or over 25 years of age, those who were studying in the evening, students who did not have to take the final degree exam and those who had visual health problems.

Instruments

This study used the virtual reality prototype "EmotionLab", built with Unreal Engine 5 and used with HMD glasses (Oculus Quest 2) with an immersive environment of 10 to 15 minutes duration, designed for emotional regulation and allowing to face two different virtual environments, Classroom and Interview Room, each with two levels of difficulty, easy and difficult, differentiated by the distracting elements. During the experience, techniques for emotional regulation were offered that allowed users to improve the execution of the requested task (oral presentation of a grade exam, for example).

Key Performance Indicators (KPIs) are the way to measure the effectiveness of the designed prototype. These KPIs were selected according to the objective of the research and the findings to be

presented to the development team at the end of the research. These were the User Experience Questionnaire (UEQ) short version, the Net Promoter Score, Time Spent on Task, Error Rate and Abandonment Rate, which are described in the following paragraphs.

The short version of the UEQ questionnaire evaluates the user's impact and perception of the properties of an interactive product. In this version, items 1 to 4 evaluate the pragmatic quality (usability) and items 5 to 8 measure the hedonic quality (user experience). It uses a 7-point rating scale to measure user experience in different dimensions. This scale ranges from -3 to +3, where -3 indicates a very negative rating. A value of 0 represents a neutral rating and +3 indicates a very positive rating.

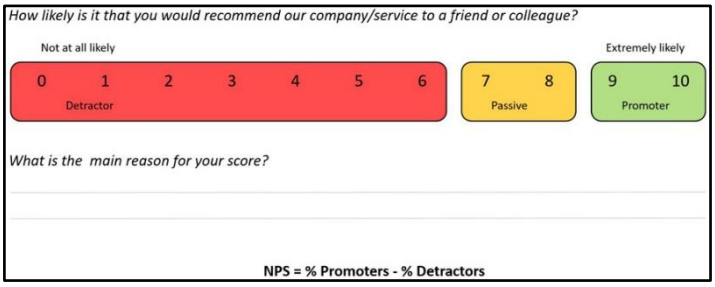
Figure 1. Items and qualities in Spanish of the UEQ-S questionnaire, extracted from the Spanish version of the UEQ

Cualidad Prágmatica	1	obstructivo	o o o o o o o	impulsor de apoyo
	2	complicado	o o o o o o o	fácil
	3	ineficiente	o o o o o o o	eficiente
	4	confuso	o o o o o o o	claro
Cualidad Hedónica	5	aburrido	o o o o o o o	emocionante
	6	no interestante	o o o o o o o	interesante
	7	convencional	o o o o o o o	original
	8	convencional	o o o o o o o	novedoso

Note: Based on (Rauschenberger et al. 2013).

The Net Promoter Score (NPS) is a measure of how loyal users will be to your product or service, measuring the likelihood that a user will recommend your product to a friend or colleague. Participants rate the question, "Would you recommend this product to a friend or colleague?" on a scale of 0 to 10.

Figure 2. Net Promoter Score NPS (Adams et al. 2022)



Note: Based on (Adams et al. 2022).

Time spent on task measures the time it takes a user to complete the task. A task can involve any function within the product you are designing, such as completing a form or making a purchase, in this case making the oral presentation linked to a degree defense or seminar. User error rate indicates the parts of a design that cause users to make mistakes. This KPI helps pinpoint areas where improvements to the user experience design should be made. Abandonment rates show how many users abandon the experience, this KPI reveals how many users abandon before reaching the end of an experience. Users may stop using the product if the navigation is difficult to understand, if they get frustrated trying to complete a task, or if they simply get bored.

Procedures

A moderated usability and user experience study was conducted. In this first experience the researcher assisted the user while performing the task and collected feedback on how he/she interacts with the designed virtual reality prototype.

Before starting the virtual reality experience, the user was told in an introductory way what the usability and user experience test consisted of, his consent was requested and any doubt was answered at the time, and finally it was indicated how the data were collected and analyzed. Once the experience was explained, the user proceeded to use the designed prototype for 10 to 15 minutes.

After using the "EmotionLab" experience, the user answered the KPIs, which is the way to measure the effectiveness of the designed prototype. These KPIs were selected according to the objective of the research and the findings to be presented to the development team once the research was completed. These were the User Experience Questionnaire (UEQ) short version, the Net Promoter Score, time spent on the task, error rate and abandonment rate.

Information analysis techniques

Statistical analysis of usability and user experience data was critical to understanding how users interact with the designed virtual reality prototype. This analysis allowed us to identify patterns, measure the effectiveness of the interface, and determine areas for improvement. The process began with data collection through methods such as surveys, usability testing, and analysis of interaction metrics. In addition, data can include both qualitative and quantitative variables, such as user satisfaction, task time, error rate, and abandonment rate

The variables listed below refer to the functionality of the virtual reality prototype "EmotionLab" and its impact on the level of usability and user experience. Thus by relating the variables it is possible to determine if the designed prototype satisfies the users and their expectations. The dependent variable was the level of usability and user satisfaction and the independent variable was the use of the designed virtual reality prototype. In this sense, the independent variable was the use of the virtual reality prototype designed in this research, which is the characteristic that is supposed to be the cause of the phenomenon studied. In turn, the dependent variable corresponded to the level of user satisfaction, task time, error rate and dropout rate.

Finally, the data were analyzed for normality with the Komogorov-Smirnov or Shapiro-wilk test. For inferential analysis, the Persons or Sperman correlation coefficient was used depending on

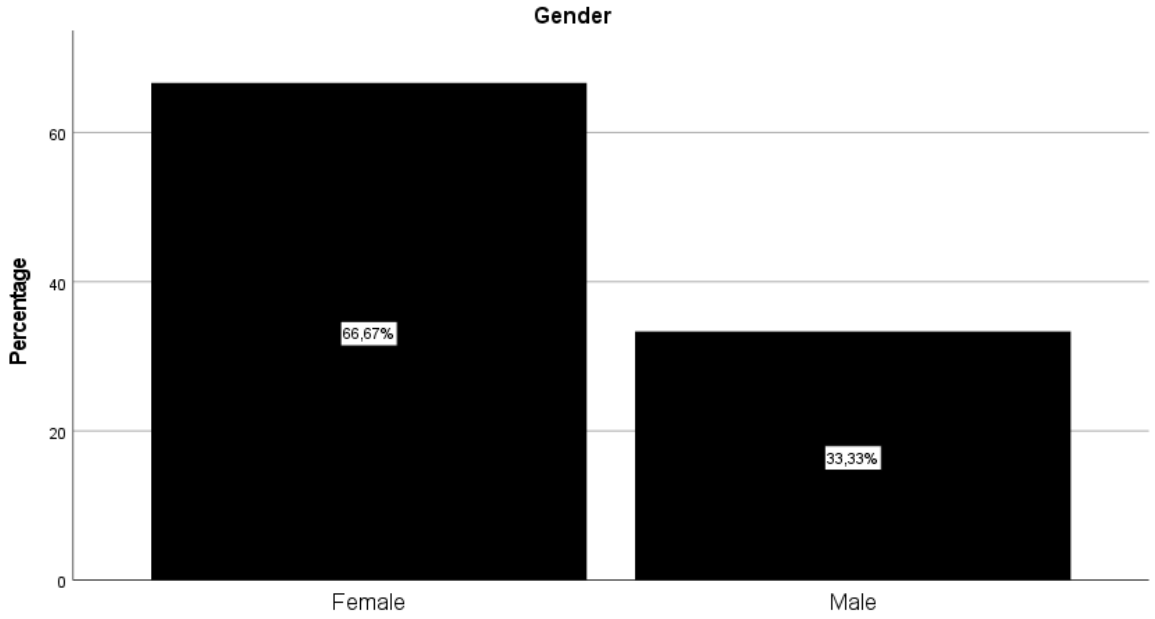
the distribution found for each variable. To establish differences, the Student's t-test or Wilcoxon test was used depending on the distribution found for each variable. Finally, a finding was considered statistically significant when the value of p<0.05

Results

Subsequent to the statistical analysis of the data collected, the results obtained considering the general and specific objectives are presented.

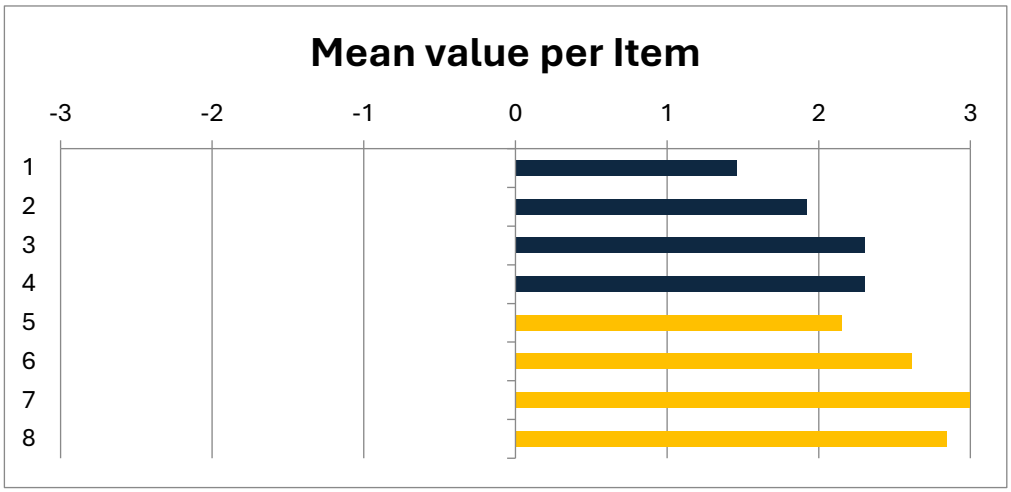
First, we describe the data of the study group, which was composed of a n=12 students of Higher Education at DuocUC, Viña del Mar campus, of which 8 were women and 4 corresponded to men, with a distribution of 66.67% female and 33.33% male and a median age of 22.5 years (Figure 3)

Figure 3. Distribution of participants by gender



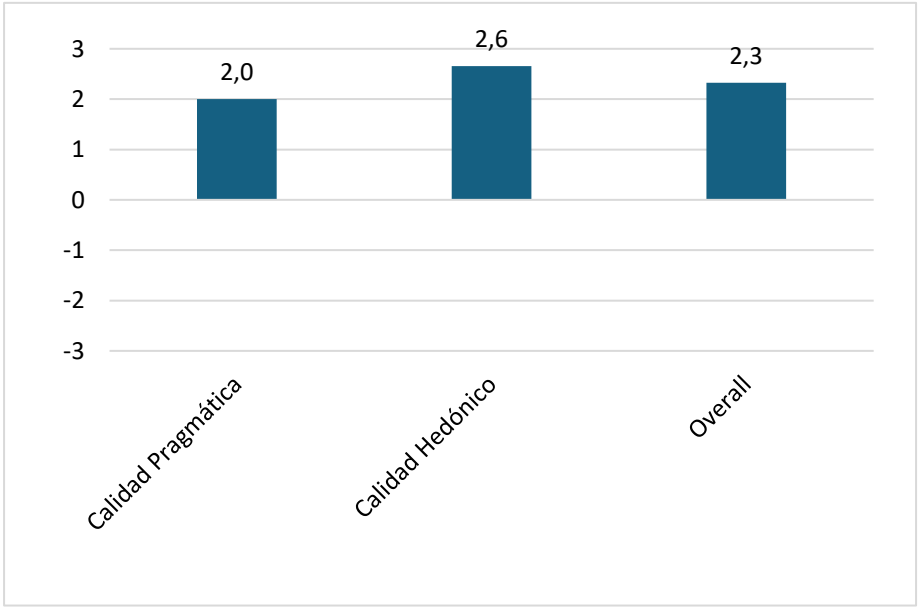
Next, with respect to usability and user experience results, the short version of the UEQ questionnaire was used to measure both classic usability aspects (efficiency, clarity, reliability) and user experience aspects (originality, stimulation). The results obtained (Figure 4) reflect a predominantly positive user experience, with high scores on key dimensions such as usability and overall user satisfaction. These findings suggest that the EmotionLab virtual reality prototype has significant potential to enhance the educational experience by providing an interactive and engaging environment for students.

Figure 4. Average values per item (UEQ-S Questionnaire)



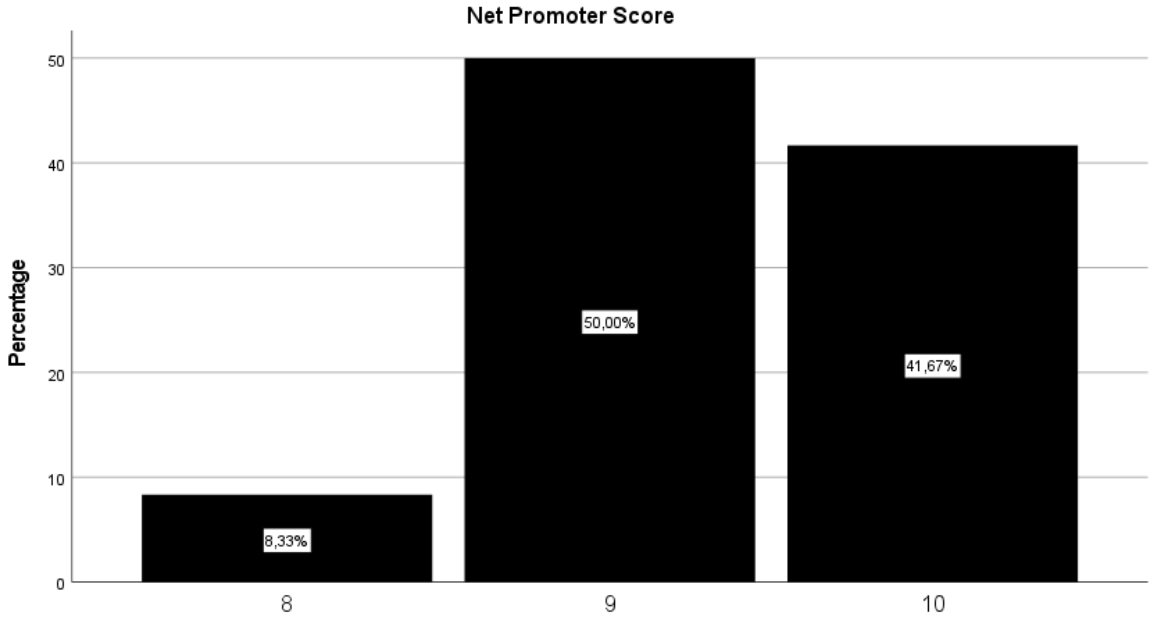
These results highlight the potential of the virtual reality prototype designed to provide an engaging and effective educational experience, enhancing both usability and student enjoyment. In relation to the pragmatic quality (Usability), a mean of 2.0 was obtained, suggesting that users find the virtual reality prototype easy to use and efficient. Likewise, the hedonic quality (User experience) reached a mean of 2.64 reflecting that users find the experience exciting, interesting and novel (Figure 5). Overall, the overall rating was 2.32 highlighting the potential of the virtual reality prototype to enhance the educational experience by providing an engaging and effective environment for students.

Figure 5. Average Values by Quality (UEQ-S Questionnaire)



Regarding the Net Promoter Score (NPS) the following was obtained, Promoters (9-10) were 11 responses (91.67%), Passives (7-8) was 1 response (8.33%) and Detractors (0-6) 0 responses (0%) obtaining an NPS applying the formula (% of Promoters - % of Detractors) of 91.67, which indicates a very high level of satisfaction and loyalty among the users of the EmotionLab virtual reality prototype. This result suggests that most users are extremely satisfied and willing to recommend the product to others.

Figure 6. Net Promoter Score



In addition, no student dropped out of the presentation during the experience. Also, none of the participants reported signs of cybersickness during the use of the experience, nor after the 15 minutes of exposure to the virtual reality environment.

Finally, the experience in the EmotionLab prototype revealed an environment where a sense of calm prevailed among the participants, reflecting the positive impact of the suggested strategies. Study participants found the recommendation to slow down when speaking useful, perceiving it as an effective tool to project calmness and control emotions such as fear, embarrassment or sadness. This inclination toward calmness as the predominant emotion underscores the tool's ability to facilitate a more serene approach to potentially stressful situations, helping to redirect stress toward more balanced and desirable emotional states such as calm and joy.

After the experience, the prototype proved to be a welcome resource for regulating emotions during complex academic situations. Users described the experience as realistic, effective and especially valuable for managing emotions in live presentation contexts, with student 1 stating that "I would recommend the use of the prototype, since, being a personal space, it has a friendly presentation". Immediate feedback emerged as a key element, highlighted by many as an essential

component that allowed them to adjust and improve their skills in real time, in this sense, student 2 stated that "I would recommend it, because it is a good way to be exposed to situations, without the need to be embarrassed with real people". This aspect seems to have left a significant impression, positioning itself as one of the strongest points of the study, with student 3 stating that "yes, I recommend it because it helps to put oneself in the context of the situation and prepares for a real exposure experience".

In terms of visual design, although the majority rated the texts and colors of the application positively, a minority expressed difficulties, suggesting an opportunity to optimize this dimension and ensure a fully accessible visual experience. Despite this, users as a whole recognized the tool's potential to strengthen their confidence in their communication skills and would highly recommend it to others. These findings highlight not only the functionality and effectiveness of the tool, but also its potential to continue to evolve and provide comprehensive support for emotional management during complex academic situations.

Discussion

The results obtained in this study underline the significant potential of the EmotionLab virtual reality prototype as an educational tool. The high rating on the pragmatic quality (mean of 2.0) indicates that users find the virtual reality prototype easy to use and efficient, which is crucial for its adoption in educational settings. The hedonic quality (mean of 2.64) reflects that users find the experience not only functional, but also exciting, interesting and novel, which can increase students' motivation and engagement, these results are critical to achieve the goal of the experience. In this sense, in terms of user experience and agreeing with this indicator is considered a key factor, however, usability, which is also relieved by these authors and has a direct relationship with the experience, is not presented in most research on serious immersive virtual reality games to enhance learning and training.

Similarly, (Ferguson et al., 2020) indicates that elements such as narrative, immersion and interaction positively influence usability and user experience, which provides positive information regarding the design of the experience. Finally, in this line of related findings, authors such as (Neher et al., 2024; Rojas-Sánchez et al., 2023) mention that usability and user experience are crucial aspects in the adoption of virtual reality simulations in educational environments, highlighting that the perception of ease of use and perceived usefulness are determining factors in the acceptance of these technologies by users, so when designing the experience in virtual reality learning environments, it must be carefully designed to maximize the effectiveness of learning and user satisfaction, which involves considering aspects such as interface, interaction and immersion.

Around the high Net Promoter Score (NPS) value of 91.67 suggests an exceptional level of satisfaction and loyalty among users, indicating that most are willing to recommend the EmotionLab prototype to other students. This level of acceptance is a positive indicator that the virtual reality prototype can be a valuable tool for enhancing the educational experience. However, although NPS is widely used, there is scant information in the scientific literature, but some studies (Adams et al. 2022) have identified benefits associated with the use of NPS, such as ease of use, high completion rates, and good comprehension by a variety of subjects, which is consistent with the findings of the present research.

In addition, the absence of cybersickness and the perception of the experience as innovative and entertaining reinforce the viability of the prototype for use with a given protocol of use in educational contexts (Ammann-Reiffer et al., 2022), , which is important to consider as stated by (Rojas-Sánchez et al., 2023) where it is pointed out that cybersickness can be a negative side effect of immersion in virtual reality environments, affecting user comfort and satisfaction during the use of this technology. This phenomenon can influence the effectiveness of learning, as users who experience dizziness or discomfort may have difficulty concentrating and fully participating in educational activities (Jaenes et al., 2024). The ability of the EmotionLab virtual reality prototype to help students identify, recognize, and regulate their emotions during complex situations such as an oral presentation is a particularly relevant challenge, as it suggests practical applications in the development of soft skills.

Conclusion

In summary, the findings obtained in this study not only validate the usability and positive user experience with the EmotionLab virtual reality prototype, but also the acceptance by students, demonstrating that the prototype has great potential to transform higher education by providing a more interactive and engaging learning environment. In this regard, future studies should consider larger and more diverse samples to confirm these results and further explore the long-term impacts, with a defined usage protocol to help students identify, recognize, and regulate their emotions during complex situations. In addition, it would be beneficial to investigate how this tool could be adapted to different educational contexts and age groups, exploring its applicability in various fields of knowledge beyond emotional education.

Similarly, the integration of complementary technologies, such as artificial intelligence (AI), could be considered to further personalize the learning experience and enhance the system's ability to respond to individual user needs. These potential enhancements could significantly increase the impact and versatility of the designed prototype in the educational setting. Incorporating progress tracking and data analysis features could provide valuable insights into students' emotional development over time. Finally, exploring the implementation of this technology in university settings could offer interesting insights into how emotional intelligence impacts the coexistence and academic performance of higher education students. Specific modules could be developed that address the emotional challenges typical of this educational stage, such as stress management, adapting to new social environments, and making important decisions for future careers. Finally, integrating EmotionLab into college counseling and tutoring programs could provide valuable additional support for students' emotional well-being and academic success.

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